

Safety Procedures

1. General safety

Before doing ANYTHING with the robot, make sure your workspace is safe.

- The workspace is clean, dry, and well-lit.
- No liquids (water, coffee, drinks) are near the robot or the battery.
- No flammable materials are near the robot or the battery.
- You know where the nearest exit is.
- You are not wearing loose jewelry or anything that could catch in the wheels.
- You have read this entire document at least once.

2. Pre-test checklist - before powering on the robot

Visual inspection

- The chassis is intact, no cracks, no loose screws.
- All three wheels spin freely by hand and are firmly attached.
- Check the wires, no wires are hanging loose where they could catch on a wheel.
- All connectors (motor, battery, BMS) are fully seated.
- The ESP32 board is firmly mounted, no loose USB cable left plugged in.
- The L298N motor driver is not visibly burnt or discolored.
- The BMS is connected correctly to the LiPo (check polarity: red = +, black = -).

Battery check

- The LiPo is NOT deformed.
- The LiPo case is intact.
- The temperature of the LiPo is correct.
- Battery voltage is within safe operating range. Do not use a LiPo below the minimum voltage per cell.
- The XT60 / battery connector is clean and undamaged.

Environment

- Test surface is flat, clear of obstacles, and large enough for the robot to move.

3. Testing the robot - safe operation

1. Place the robot on the test surface, wheels down in the sphere.
2. Connect the battery to the BMS / power input. Listen for any unusual sound. If you hear anything strange, disconnect immediately.
3. Check that the ESP32 LED comes on as expected. If nothing lights up, disconnect and inspect.
4. Connect to the ESP32 (Wi-Fi / Bluetooth) from the control interface.

5. Start with the LOWEST motor speed. Send a short test command (e.g., forward 1 second).
6. If the robot behaves erratically, spins violently, or moves in the wrong direction, send a STOP command.
7. Gradually increase speed only after confirming the robot responds correctly.
8. Never leave the robot powered on and unattended.

STOP IMMEDIATELY if you notice:

Smoke • Burning smell • Unusual heat from the battery or boards • Crackling/popping sounds • The LiPo starts to swell • The robot accelerates uncontrollably • Any wire glowing or melting. Action: Disconnect the battery FIRST, then move the robot away from flammable materials.

4. Shutdown procedure - after using the robot

1. Send a STOP command from the control interface.
2. Wait for the sphere to come to a complete stop.
3. Disconnect the battery from the BMS / power input.
4. Inspect the LiPo visually for any swelling or damage. If it is hot, do NOT charge it immediately.
5. Make sure the robot is in good condition (cables, wheels, chassis).
6. Place the robot back in its storage area.
7. Store the LiPo separately.

5. LiPo battery - before charging

Check the LiPo guide document.

6. Storage - when not in use

- Battery is removed from the robot before long-term storage.
- Robot is stored in a dry place, away from dust and impacts.
- Cables are tidy and not under stress.
- All boards are protected from accidental short-circuits.

7. Emergency procedures - what to do if something goes wrong

Robot is out of control

1. Stay calm. Do not try to grab a spinning robot.
2. Send a STOP command from the control interface.
3. If it does not respond, physically disconnect the battery (pull the main connector).
4. Inspect the robot before powering it on again.

Smoke or burning smell

1. Disconnect the battery immediately.
2. Move the robot to a non-flammable surface, away from people and flammables.
3. Ventilate the room (open windows, leave the door open). The fumes from a burning LiPo are toxic.
4. Do not breathe in the smoke.
5. If smoke continues or you see flames, follow the fire procedure below.